

Week 1 Task DevOps intern

Task1: Ubuntu Installation

Installed Ubuntu using vagrant

```
> vagrant up
Bringing machine 'ubuntu' up with 'virtualbox' provider...
==> ubuntu: Importing base box 'bento/ubuntu-24.04'...
==> ubuntu: Matching MAC address for NAT networking...
==> ubuntu: Checking if box 'bento/ubuntu-24.04' version '202502.21.0' is up to date...
==> ubuntu: Setting the name of the VM: ubuntu_vm
==> ubuntu: Clearing any previously set network interfaces...
==> ubuntu: Preparing network interfaces based on configuration...
ubuntu: Adapter 1: nat
ubuntu: Adapter 2: hostonly
==> ubuntu: Forwarding ports...
ubuntu: 22 (guest) => 2222 (host) (adapter 1)
==> ubuntu: Running 'pre-boot' VM customizations...
==> ubuntu: Booting VM...
==> ubuntu: Waiting for machine to boot. This may take a few minutes...
ubuntu: SSH address: 127.0.0.1:2222
ubuntu: SSH username: vagrant
ubuntu: SSH auth method: private key
ubuntu:
ubuntu: Vagrant insecure key detected. Vagrant will automatically replace
ubuntu: this with a newly generated keypair for better security.
ubuntu:
ubuntu: Inserting generated public key within guest...
ubuntu: Removing insecure key from the guest if it's present...
ubuntu: Key inserted! Disconnecting and reconnecting using new SSH key...
==> ubuntu: Machine booted and ready!
==> ubuntu: Checking for guest additions in VM...
ubuntu: The guest additions on this VM do not match the installed version of
ubuntu: VirtualBox! In most cases this is fine, but in rare cases it can
ubuntu: prevent things such as shared folders from working properly. If you see
ubuntu: shared folder errors, please make sure the guest additions within the
ubuntu: virtual machine match the version of VirtualBox you have installed on
ubuntu: your host and reload your VM.
ubuntu:
ubuntu: Guest Additions Version: 7.1.6
ubuntu: VirtualBox Version: 7.2
==> ubuntu: Setting hostname...
==> ubuntu: Configuring and enabling network interfaces...
==> ubuntu: Mounting shared folders...
ubuntu: /home/horrorondor/Downloads => /vagrant
```

```
> vagrant ssh
Welcome to Ubuntu 24.04.2 LTS (GNU/Linux 6.8.0-53-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Wed Jul 29 06:28:05 AM UTC 2026

System load:          0.27
Usage of /:            14.0% of 30.34GB
Memory usage:         9%
Swap usage:           0%
Processes:            160
Users logged in:      0
IPv4 address for eth0: 10.0.2.15
IPv6 address for eth0: fd17:625c:f037:2:a00:27ff:fe6b:69c9

This system is built by the Bento project by Chef Software
More information can be found at https://github.com/chef/bento

Use of this system is acceptance of the OS vendor EULA and License Agreements.
vagrant@ubuntu:~$ lsb_release -a
No LSB modules are available.
Distributor ID: Ubuntu
Description:    Ubuntu 24.04.2 LTS
Release:       24.04
Codename:      noble
```

fig 1: Screenshot of ubuntu installation and linux terminal

Task 2: Create users and Groups

```
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nologin
systemd-timesync:x:997:997:systemd Time Synchronization:/:/usr/sbin/nologin
dhcpcd:x:100:65534:DHCP Client Daemon,,,:/usr/lib/dhcpcd:/bin/false
messagebus:x:101:102::/nonexistent:/usr/sbin/nologin
systemd-resolve:x:992:992:systemd Resolver:/:/usr/sbin/nologin
pollinate:x:102:1::/var/cache/pollinate:/bin/false
polkitd:x:991:991:User for polkitd:/:/usr/sbin/nologin
syslog:x:103:104::/nonexistent:/usr/sbin/nologin
uidd:x:104:105::/run/uidd:/usr/sbin/nologin
tcpdump:x:105:107::/nonexistent:/usr/sbin/nologin
tss:x:106:108:TPM software stack,,,:/var/lib/tpm:/bin/false
landscape:x:107:109::/var/lib/landscape:/usr/sbin/nologin
fwupd-refresh:x:989:989:Firmware update daemon:/var/lib/fwupd:/usr/sbin/nologin
usbmux:x:108:46:usbmux daemon,,,:/var/lib/usbmux:/usr/sbin/nologin
sshd:x:109:65534::/run/sshd:/usr/sbin/nologin
vagrant:x:1000:1000:vagrant:/home/vagrant:/bin/bash
vboxadd:x:999:1::/var/run/vboxadd:/bin/false
user1:x:1001:1002::/home/user1:/bin/bash
user2:x:1002:1004::/home/user2:/bin/sh
```

Fig 2.1: User lists

```
vboxsf:x:988:
vboxdrmipc:x:987:
group1:x:1001:user1
user1:x:1002:
group2:x:1003:user2
user2:x:1004:
vagrant@ubuntu:/etc$
```

Fig 2.2: Group list

Task3: File Permission

```
vagrant@ubuntu:~$ ls -al
total 32
drwxr-x--- 4 vagrant vagrant 4096 Feb 20 2025 .
drwxr-xr-x 5 root     root    4096 Jul 29 07:00 ..
-rw-r--r-- 1 vagrant vagrant  220 Mar 31 2024 .bash_logout
-rw-r--r-- 1 vagrant vagrant 3771 Mar 31 2024 .bashrc
drwx----- 2 vagrant vagrant 4096 Feb 20 2025 .cache
-rw-r--r-- 1 vagrant vagrant  807 Mar 31 2024 .profile
drwx----- 2 vagrant vagrant 4096 Jul 29 06:26 .ssh
-rw-r--r-- 1 vagrant vagrant    0 Feb 20 2025 .sudo_as_admin_successful
-rw-r--r-- 1 vagrant vagrant    5 Feb 20 2025 .vbox_version
vagrant@ubuntu:~$
```

Fig 3.1: File permission

Task4: Networking commands

```
vagrant@ubuntu:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:6b:69:c9 brd ff:ff:ff:ff:ff:ff
    altname enp0s3
    inet 10.0.2.15/24 metric 100 brd 10.0.2.255 scope global dynamic eth0
        valid_lft 82454sec preferred_lft 82454sec
    inet6 fd17:625c:f037:2:a00:27ff:fe6b:69c9/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 86211sec preferred_lft 14211sec
    inet6 fe80::a00:27ff:fe6b:69c9/64 scope link
        valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:fd:84:64 brd ff:ff:ff:ff:ff:ff
    altname enp0s8
    inet 192.168.56.13/24 brd 192.168.56.255 scope global eth1
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fe6b:8464/64 scope link
        valid_lft forever preferred_lft forever
```

Fig 4.1: ip address show

```

vagrant@ubuntu:~$ nslookup -debug youtube.com
Server:          127.0.0.53
Address:         127.0.0.53#53

-----
      QUESTIONS:
        youtube.com, type = A, class = IN
      ANSWERS:
        -> youtube.com
            internet address = 142.251.222.206
            ttl = 226
      AUTHORITY RECORDS:
      ADDITIONAL RECORDS:
-----
Non-authoritative answer:
Name:   youtube.com
Address: 142.251.222.206
-----
      QUESTIONS:
        youtube.com, type = AAAA, class = IN
      ANSWERS:
        -> youtube.com
            has AAAA address 2404:6800:4007:836::200e
            ttl = 1
      AUTHORITY RECORDS:
      ADDITIONAL RECORDS:
-----
Name:   youtube.com
Address: 2404:6800:4007:836::200e

```

fig 4.2: Nameserver lookup

```

vagrant@ubuntu:~$ ss -tulnp

```

Netid	State	Recv-Q	Send-Q	Local Address:Port	Peer Address:Port	Process
udp	UNCONN	0	0	127.0.0.54:53	0.0.0.0:*	
udp	UNCONN	0	0	127.0.0.53%lo:53	0.0.0.0:*	
udp	UNCONN	0	0	10.0.2.15%eth0:68	0.0.0.0:*	
tcp	LISTEN	0	4096	127.0.0.53%lo:53	0.0.0.0:*	
tcp	LISTEN	0	4096	127.0.0.54:53	0.0.0.0:*	
tcp	LISTEN	0	4096	*:22	*:*	

fig 4.2: Socket statistics